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# Transverse and longitudinal coefficient of thermal expansion of carbon fibers at high temperatures (300–2500 K)

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## ABSTRACT

The transverse and longitudinal coefficients of thermal expansion (CTE) of carbon fibers are important parameters in C/C composites behaviour. In this study, the measurements of CTE are presented at very high temperatures (up to 2500 K). Experimental validations are first realised on isotropic metallic (tungsten) fiber. Then, measurements are performed on four carbon fibers (rayon-based, PAN-based and pitch-based). The results show that the mean transverse CTE varies from  $5 \times 10^{-6} \text{ K}^{-1}$  to  $10 \times 10^{-6} \text{ K}^{-1}$  and the longitudinal one from  $1.6 \times 10^{-6} \text{ K}^{-1}$  to  $2.1 \times 10^{-6} \text{ K}^{-1}$  depending on the fiber structural properties. For that, the relationships between the structural properties and the heat treatment temperature of the carbon fibers are discussed in relation to thermal expansion behaviour. A calculation of the ratio between the transverse and the longitudinal measured CTE as a function of the temperature allows us to classify fibers from the most isotropic to the most anisotropic in relation with the structural properties measured at room temperature.

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## 1. Introduction

Carbon fibers present very efficient properties at very high temperatures: wide range of Young's modulus, thermal conductivity and thermal expansion [1,2]. These excellent properties are due to the ordered structure, close to that of single crystal graphite, created during melt spinning and perfected during oxidation and graphitization. At very high temperatures [3,4], the literature gives very little data on the properties of the carbon fibers. This study focuses on transverse and longitudinal CTE measurements at very high temperatures, it is included in a large program devoted to mechanical [5], thermal [6,7] and thermomechanical characterisation of carbon fibers.

The difficulties of such measurements are due to the micro scale of the fibers ( $\approx 10 \mu\text{m}$ ), the wide range of temperatures (300–2700 K) and to the small variations of the fiber diameter ( $\Delta d \approx 1\text{--}30 \text{ nm}$ ) and length ( $\Delta L \approx 1\text{--}50 \mu\text{m}$ ). An exper-

imental device and a data processing method was previously developed for transverse [8] and longitudinal [9] CTE measurements. They allow characterisation with extreme accuracy, sensitivity of 300 pm on diameter variation,  $0.1 \mu\text{m}$  on length variation and measurements uncertainties lower than  $\pm 5\%$  for both of them.

The first aim of this study is the presentation of transverse and longitudinal measurements on various kinds of fibers at a high temperature. Then, the calculation of the ratio between transverse and longitudinal coefficients of thermal expansion will be used to estimate the anisotropy of the fibers with increasing temperature. By comparison with fibers structural properties the question of how the CTE will vary according to the fibers textural properties and this will be discussed. Finally, we will set out a classification of fibers in order to establish that such measurements at very high temperatures improve the understanding of carbon fibers thermomechanical behaviour and structural properties.

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## 2. Brief description of the CTE measurement methods

### 2.1. Definition of the transverse and longitudinal CTE

Three kinds of CTE are available in the literature (see ref. [10]). In the following definitions, parameter  $\beta$  means either the fiber diameter  $d$  or the fiber length  $L$ . These CTEs are defined as the specific CTE in Eq. (1), the average one in Eq. (2) and the real one Eq. (3):

$$\alpha_s(T) = \frac{\beta(T) - \beta(T_0)}{\beta(T_0)} \quad (1)$$

$$\alpha_m(T) = \frac{1}{T - T_0} \frac{\beta(T) - \beta(T_0)}{\beta(T_0)} \quad (2)$$

$$\alpha_T(T) = \frac{1}{\beta(T_0)} \frac{\partial \beta(T)}{\partial T} \quad (3)$$

All those CTEs depend on according to the room temperature  $T_0$ . The experimental results presented later concerned the specific CTE. Such definition Eq. (1) is similar to a transverse or longitudinal deformation of the fiber.

### 2.2. Measurements of the Longitudinal CTE

Very little data is available at high temperatures on the thermal expansion of carbon fibers [4,11,12]. Sauder et al. [9], developed the technique used here. A very low stress is applied onto the fiber of 50 mm length. This corresponds to 0.01% of its maximum longitudinal elongation measured at room temperature. Constant stress measured by sensor force is maintained during fiber heating through control of grip displacement. The applied displacement balances the longitudinal expansion of the fiber induced by heating. Such thermal expansion measurements are very easy and accurate with computer data processing.

### 2.3. Measurements of the transverse CTE

Two standard practices are available to measure the transverse CTE. The first group used an indirect method [13,14]. That means a global dilatation measurement of several thousands of fibers in a mini-composite. The second group [12,15–18] used a direct method performed on a single fiber. Here, the authors performed fiber diameter measurements at increasing temperatures using a laser diffraction technique. Regardless of the method used, micro scale and weak diameter variations ( $\approx 10$  nm) involved rather delicate measurements.

In previous work, a method for transverse CTE estimation was elaborated at very high temperatures. Even if optical methods have been previously used [15–17], one of the major points of this work [8] was the improvements in signal and data processing for parameter estimation. In this work, we present the optical method based on the laser diffraction phenomenon at a micro scale (fibers). The inverse method developed is based on reduced knowledge of the Fraunhofer's model. An analysis of different kinds of error sources (fiber roughness, fiber stress, 1D array sensor resolution) allows a validation of this work for transverse CTE measurement on carbon fiber.

## 3. Experimental details

### 3.1. Apparatus description

The experimental device (Fig. 1), was developed on a bench designed for mechanical characterisation [19,8]. The fiber of 50 mm length is heated up by Joule effect. The fiber temperature is assumed to be quasi-isotherm ( $T_{\text{center}} - T_{\text{surface}} < 2^\circ\text{C}$ , [7]). A very low pressure ( $P < 10^{-3}$  Pa) is maintained around the fiber in a silica glass enclosure. A bichromatic pyrometer measures the absolute temperature of the fiber, while two CMOS sensors of 1024 pixels (size of:  $25 \times 25 \mu\text{m}^2$ ) collect the

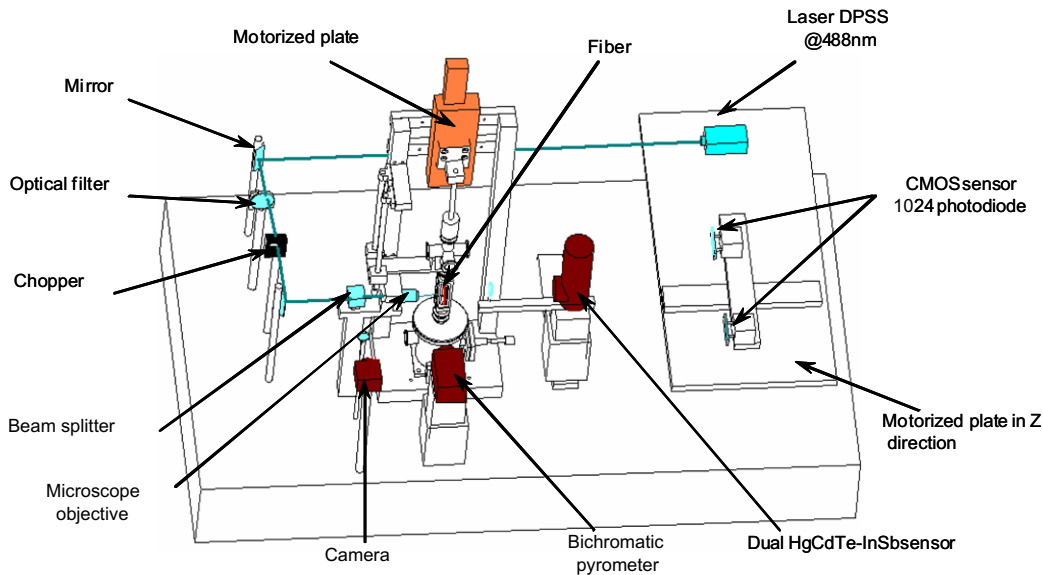


Fig. 1 – Scheme of experimental device.

**Table 1 – Main characteristics of fibers at room temperature**

| Fibers                             | XN05       | PANEX 33         | HTA 5131          | P100              |
|------------------------------------|------------|------------------|-------------------|-------------------|
| Precursors                         | Iso. pitch | PAN              | PAN               | An pitch          |
| Young's modulus (GPa) <sup>b</sup> | 53         | 230 <sup>b</sup> | 248 <sup>b</sup>  | 690 <sup>a</sup>  |
| Density <sup>c</sup>               | 1.63       | 1.81             | 1.77 <sup>a</sup> | 2.15 <sup>a</sup> |
| Mean diameter (μm) <sup>d</sup>    | 10         | 7                | 7                 | 11 <sup>a</sup>   |

a From supplier.  
b Measured by tensile tests [5].  
c Measured using a helium pycnometer.  
d Image analyses of scanning electron micrographs photography of fiber cross sections.

diffracted intensity of an argon laser ( $\lambda = 488$  nm). The diffracted patterns are visualised on a numerical oscilloscope and then transferred to a computer with an IEEE 488–GPIB card where numerical data processing is carried out.

During these measurements, fiber stress is kept at a small percentage with a traction table controlled by a force sensor HBM 5 N. The global device is automatically controlled in order to record a large quantity of data to decrease the influence of signal noise. The characterisation of a fiber from 300 to 2800 K takes 2 h.

### 3.2. Experimental routine

At room temperature  $T_0$ , the fiber is stressed at 0.01% of its maximum longitudinal elongation. The length  $L_0$  of the fiber is measured and five diffracted patterns are recorded. The temperature is increased of one-step (typically 50 K) at temperature  $T_i$ . During this phase, fiber stress is kept constant with a traction table remote controlled by a force sensor. The traction table travel  $L_i$  is recorded by a sensor of 0.1 μm displacement sensitivity [9]. In this new position, a second round of five diffracted patterns is performed and the data processing method [8] is completed.

## 4. Structural properties of the fibers

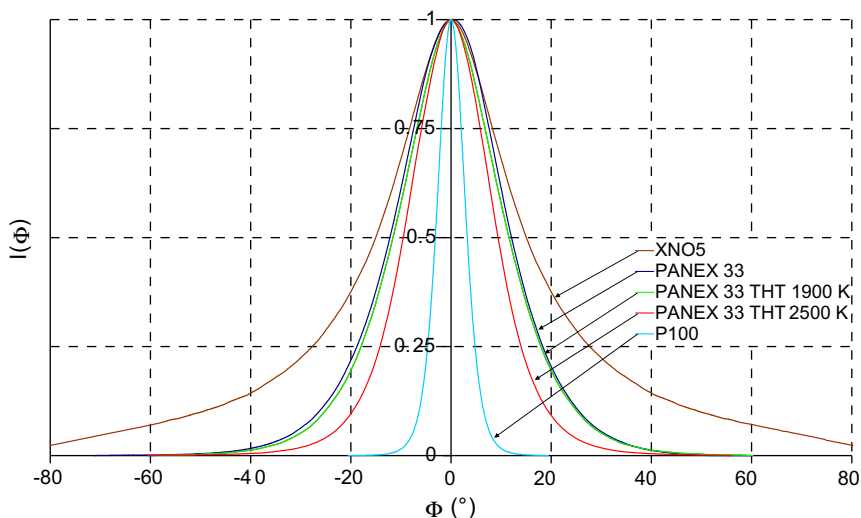
### 4.1. Description of the carbon fibers

In this study, four kinds of carbon fibers were measured: XN05, HTA 5131, PANEX 33 and P100. The XN05 fiber (Nippon Graphite Fiber Co.) is produced from an isotropic pitch precursor and heat-treated for 2 h at a temperature of 2500 K. The HTA 5131 (Tenax) and the PANEX 33 fiber (ZOLTEK Corp.) come from a PAN precursor. The PANEX 33 was heat-treated at two temperatures (1900 and 2500 K). Finally the P100 fiber (AMO-CO Performance Products Co., formerly Union Carbide Co.) come from an anisotropic pitch precursor with heat treat-

**Table 2 – Structural characteristics of fibers at room temperature**

| Fiber name | Heat treatment (K) | $d_{002}$ (nm) | $L_c$ (nm) | $Z(^{\circ})$ | Density |
|------------|--------------------|----------------|------------|---------------|---------|
| XN05       | 2500               | 0.345          | 3.5        | 30            | 1.63    |
| PANEX 33   | 1500 <sup>a</sup>  | 0.349          | 1.9        | 24.5          | 1.75    |
|            | 1900               | 0.345          | 2.4        | 23            | 1.77    |
| HTA 5131   | 2500               | 0.343          | 4.2        | 19            | 1.82    |
|            | 1400 <sup>a</sup>  | 0.347          | 2.1        | 24.1          | 1.77    |
| P100       | 2700 <sup>a</sup>  | 0.339          | 10.6       | 6.4           | 2.15    |

a As received.

**Fig. 2 – Distributions of intensity of scattering  $I(\Phi)$  for carbon fibers.**

ment at above 2700 K by the supplier. For each fiber, the highest measured temperature never goes over that of the heat treatment. Therefore, we assumed that fibers would be stable during the measurement process.

The characteristics summarized in Table 1 indicate that a wide spectrum of fibers is covered. Except for the P100 fiber, these characteristics were measured in the laboratory. For the measurements, the length of all the fibers is equal to 50 mm.

#### 4.2. Structural characterisation of the fibers

$L_c$ ,  $L_a$ , and  $d_{002}$  correspond to standard parameters used in the description of carbon structures.  $L_a$  is the longitudinal dimension of crystallites;  $L_c$  is the lateral dimension and  $d_{002}$  is the mean distance between two successive layers. These parameters were measured with X-ray diffraction on such carbon fibers.  $L_c$  and  $d_{002}$  were determined using the Scherrer equation from the positions of the diffraction

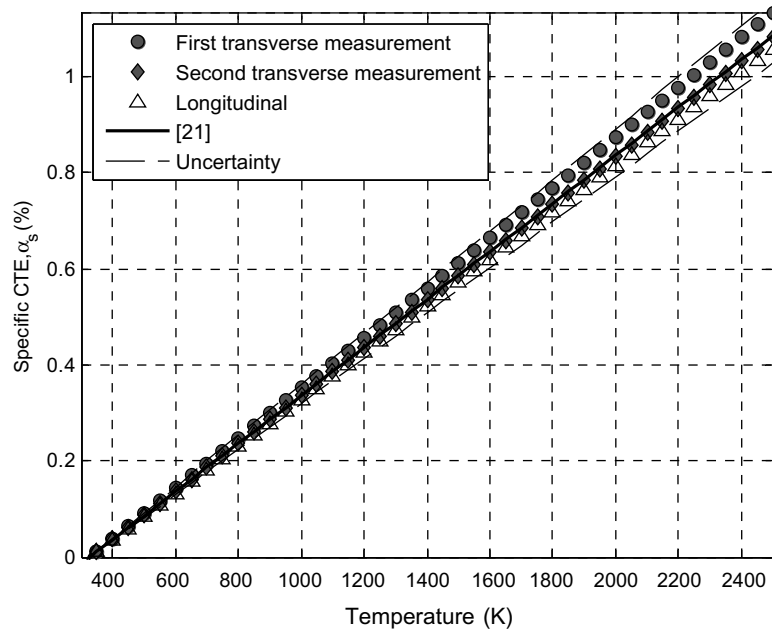


Fig. 3 – Measurement on tungsten metallic isotropic filament.

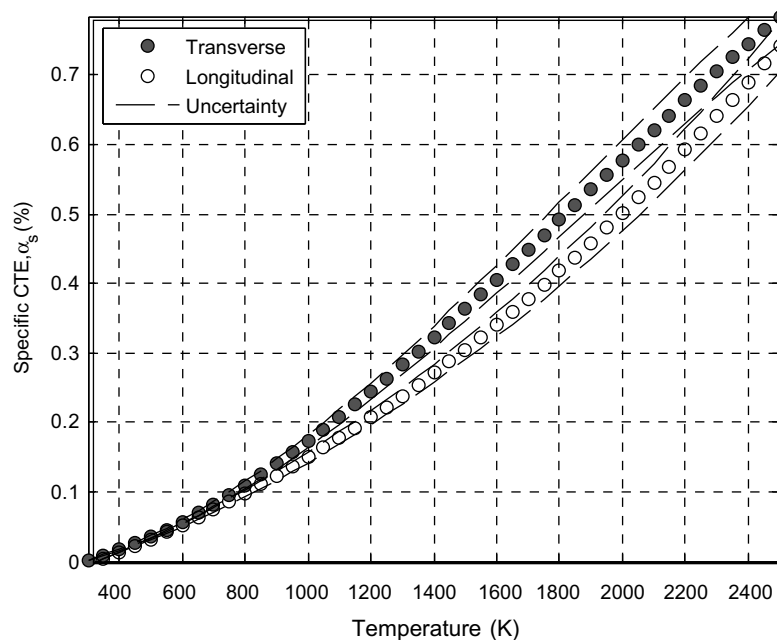


Fig. 4 – Measurement on XN05 isotropic pitch fiber.

maxima and the width at half-maximum intensity of the (002) and (110) peaks. The orientation distribution of graphitic planes was determined by X-ray analysis of the fibers, as described by Ruland [20]. It was obtained from an azimuthal spread of the 002-reflection arc. This provides an indication of the preferred orientation of the basic structural units (BSU) relative to fiber axis. The orientation

parameter  $Z$  (in degrees) is the full width at half the maximum intensity obtained in the azimuthal scan. Fiber texture was described using both the distribution of the intensity of scattering at angle  $\phi$  and the orientation parameter  $Z$ . Measured parameters describing the structure and the texture of the fibers are presented in Table 2. The fibers are ranked by increasing Young's modulus. We can

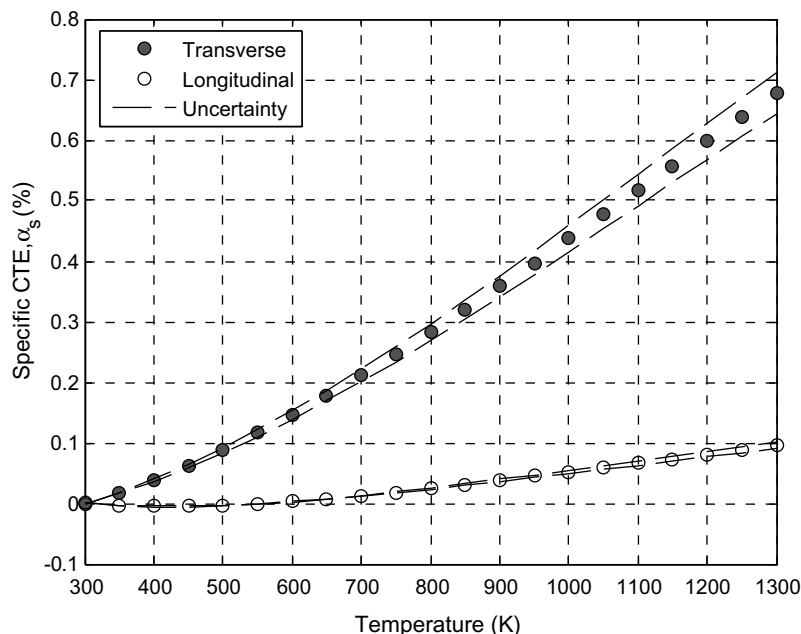


Fig. 5 – Measurement on HTA 5131 ex-PAN fiber.

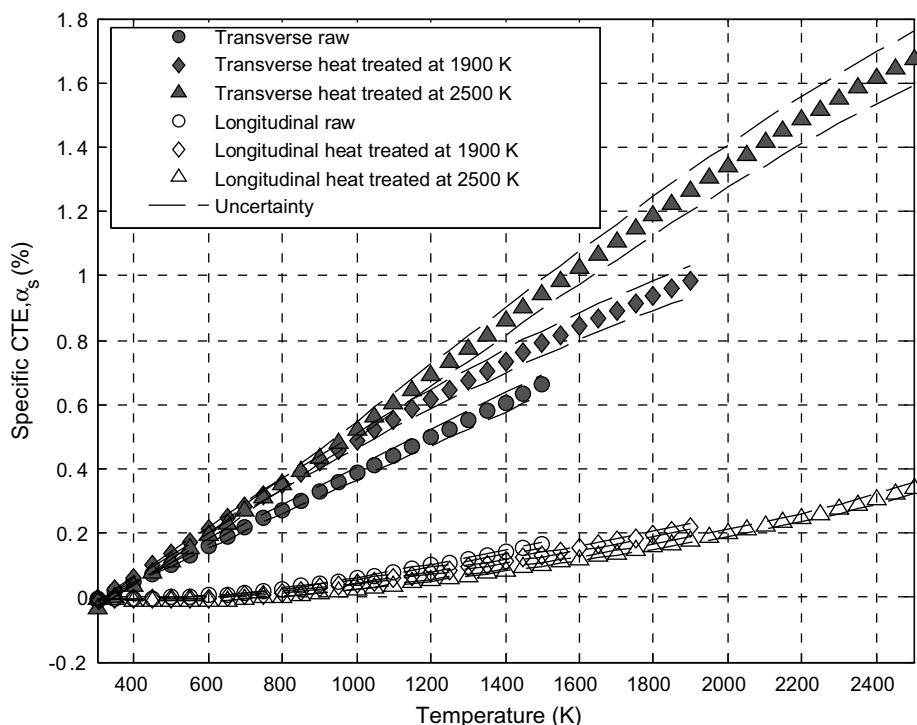


Fig. 6 – Measurement on PANEX 33 ex-PAN fibers.

notice that each parameter obviously varies with respect to fibers stiffness:  $L_c$  increases,  $d_{002}$  decreases,  $Z$  decreases and density increases.

This trend indicates the presence of crystallites with an increasing size, separated by a shorter distance between two successive layers and with a more pronounced orientation in the fiber axis direction. This reflects an increasing degree of anisotropy. It is illustrated in Fig. 2 where the distribution of intensity of scattering  $I(\phi)$  shows that fiber XN05 is the most isotropic, whereas fiber P100 is the most anisotropic.

## 5. Results and discussion

### 5.1. Validation on isotropic fibers

Validation of the transverse and longitudinal CTE measurements was performed using a metallic tungsten isotropic fiber with well-known bulk CTE [21].

The dimensions of the tungsten fiber are 18  $\mu\text{m}$  diameter and 50 mm long. The heating ramp applied to the fiber was from 300 K to 2000 K by steps of 50 K. Both transverse and longitudinal measurements were performed at each step accord-

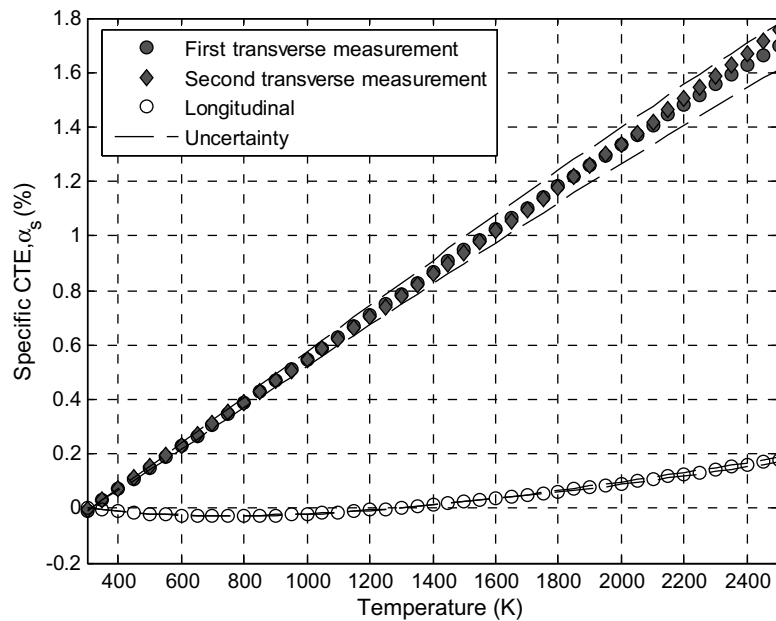


Fig. 7 – Measurement on P100 anisotropic pitch fiber.

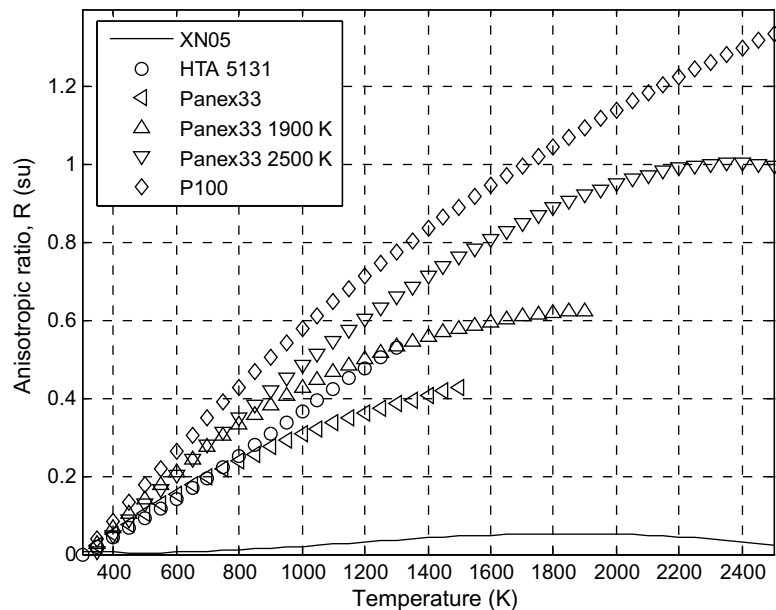


Fig. 8 – Anisotropy degree  $R$  of measured carbon fibers.

ing to the experimental routine previously described. In Fig. 3, the measured CTE profiles are compared to the bulk properties [20]. With a difference of less than 5% between the experimental data measured here and the given bulk CTE, we assumed that a very good agreement does exist. Since the tungsten fiber is isotropic, the transverse and longitudinal CTEs are equal at any temperature. The result presented

Fig. 3 confirms this assumption and allows a validation of the measurements methods.

## 5.2. Measurements of the CTE on carbon fibers

Previous results obtained on isotropic metallic fiber allow us to perform measurements and advanced material interpreta-

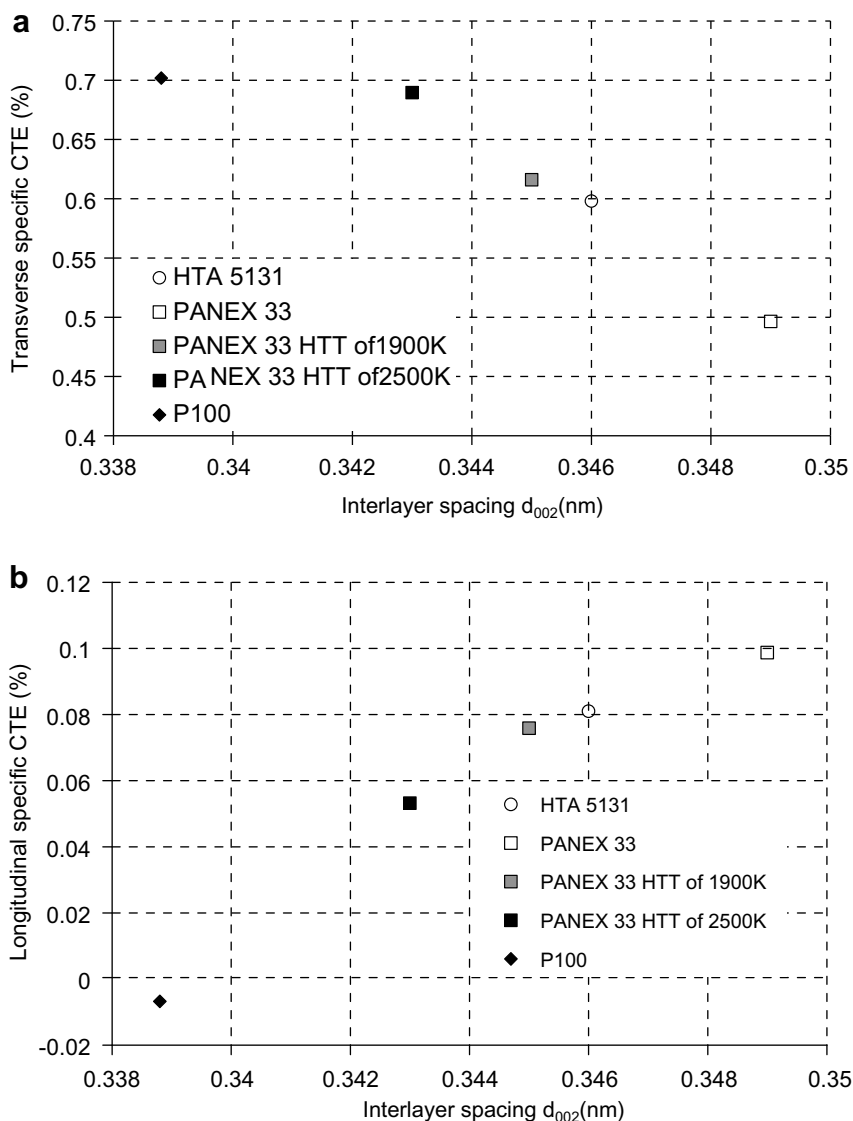


Fig. 9 – Transverse (a) and longitudinal (b) carbon fibers specific CTE dependence on the interlayer spacing at 1200 K.

Table 3 – Polynomial coefficients of measured transverse CTE

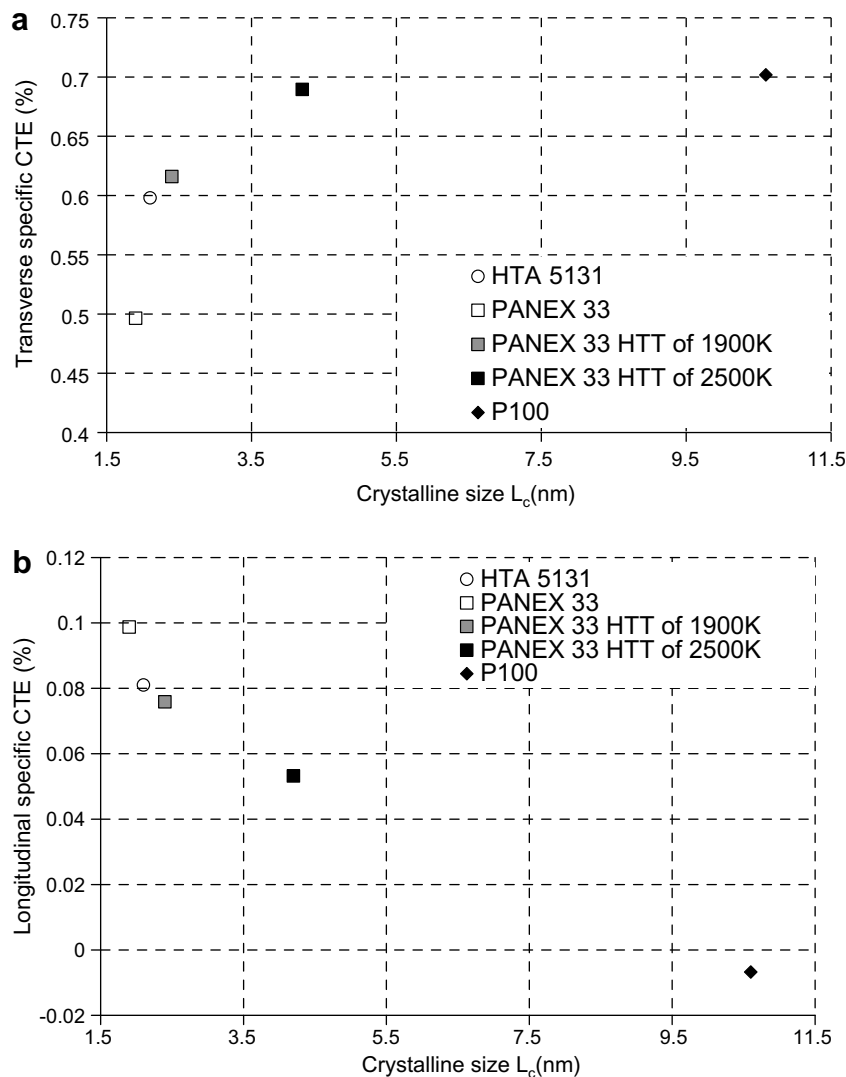
| Fiber name       | Transverse |            |            |            |
|------------------|------------|------------|------------|------------|
|                  | $a_3$      | $a_2$      | $a_1$      | $a_0$      |
| Tungsten         | –          | –          | 4.982E–04  | –1.625E–01 |
| XN05             | –4.090E–11 | 2.300E–07  | 1.780E–06  | –1.810E–02 |
| HTA 5131         | –2.315E–10 | 7.772E–07  | –6.197E–05 | –4.669E–02 |
| Panex33          | 1.059E–22  | –1.360E–08 | 5.850E–04  | –1.860E–01 |
| Panex33 (1900 K) | –2.008E–11 | –4.201E–08 | 8.026E–04  | –2.519E–01 |
| Panex33 (2500 K) | –5.201E–11 | 1.874E–07  | 6.221E–04  | –2.368E–01 |
| P100             | 1.238E–11  | –3.885E–08 | 8.222E–04  | –2.502E–01 |

**Table 4 – Polynomial coefficients of measured longitudinal CTE**

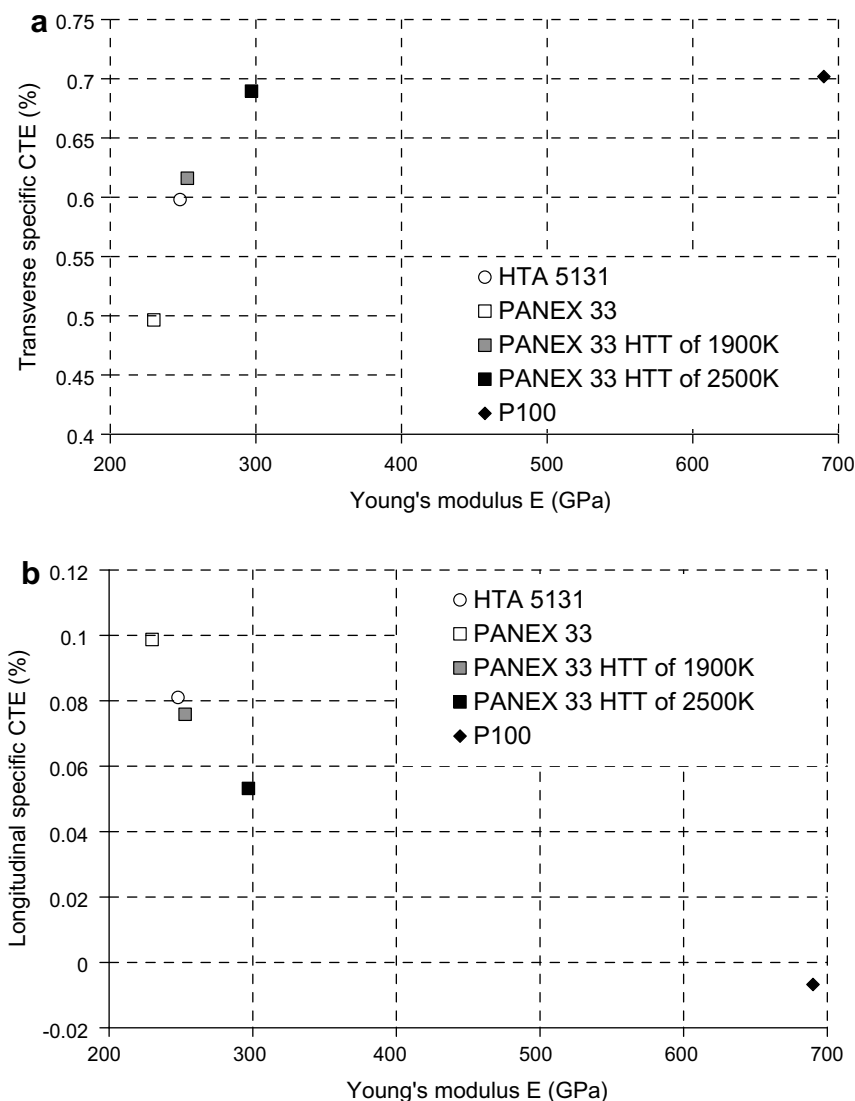
| Fiber name       | Longitudinal |            |           |            |            |
|------------------|--------------|------------|-----------|------------|------------|
|                  | $a_4$        | $a_3$      | $a_2$     | $a_1$      | $a_0$      |
| Tungsten         | –            | –          | –         | 4.874E–04  | –1.602E–01 |
| XN05             | –            | –          | 8.060E–08 | 1.123 E–04 | –4.437E–02 |
| HTA 5131         | 1.223E–13    | –5.226E–10 | 8.481E–07 | –4.753E–04 | 7.955E–02  |
| Panex33          | –1.288E–14   | –1.877E–11 | 2.228E–07 | –1.569E–04 | 2.529E–02  |
| Panex33 (1900 K) | 2.966E–14    | –1.714E–10 | 4.109E–07 | –2.742E–04 | 4.798E–02  |
| Panex33 (2500 K) | 2.808E–14    | –1.671E–10 | 4.070E–07 | –2.978E–04 | 5.506E–02  |
| P100             | 1.152E–14    | –8.617E–11 | 2.766E–07 | –2.867E–04 | 6.391E–02  |

tion on various carbon fibers. Here, the goal is to improve quantitative study linking fibers thermomechanical behaviour to structural properties. Measurements were performed on four carbon fibers described before (see Table 1). Three different specimens are measured for each type of fibers. For one specimen, 25 measurements are performed at each temperature and three heating (i.e. cooling) and reheating ramps are done. With all this data, the uncertainty and the repeatability

of the measurements was calculated. The CTE uncertainty for one specimen is always lower than 5% for the three heating stages. The deviation between three different specimens of each type of fiber is lower than 10%. For each experimental measurement, the CTE uncertainties are reported in dashed line for all the figures. The experimental measurements are presented below in ascending rank from isotropic to most anisotropic fibers.

**Fig. 10 – Transverse (a) and longitudinal (b) carbon fibers specific CTE dependence on the crystalline size at 1200 K.**





**Fig. 11 – Transverse (a) and longitudinal (b) carbon fibers specific CTE dependence on the Young's modulus at 1200 K.**

First, measurements of the transverse and longitudinal specific CTE on the heat-treated (at 2500 K) XN05 fiber are plotted (Fig. 4) as function of temperatures from 300 K to 2500 K. The specific longitudinal and the transverse CTE measured are quite close for the whole temperature range. As predicted by the structural analysis, the results show that this fiber is quasi-isotropic.

The next fiber is the HTA 5131, where the longitudinal and the transverse CTEs are plotted (Fig. 5) as functions of temperatures from 300 to 1300 K. Here a difference at any temperature between longitudinal and transverse CTEs is substantial.

The longitudinal and the transverse CTEs of the PANEX 33 fiber are plotted (Fig. 6) as functions of temperatures for different heat-treated temperatures. The measurement range goes from: (i), 300 to 1500 K for the raw fiber, (ii), 300 to 1900 K for the heat-treated fiber at 1900 K and (iii), 300 to 2500 K for the heat-treated fiber at 2500 K. Fig. 8 shows the effects of the thermal treatments on both longitudinal and transverse CTE. This is consistent with the structural properties results (Table 2). As the heat treatment tempera-

ture increases from raw to 2500 K,  $d_{002}$  decreases and  $L_c$  increases. This is due to the textural reorganisation of the fiber. However, at the same time, the mass density increases. A comparison of the transverse and the longitudinal CTE shows that the increasing of the heat-treated temperature caused the increase of the transverse CTE values whereas the longitudinal CTE decreases. This is due to the graphitic plane re-lining along the fiber longitudinal axis. This is consistent with the Young's modulus and the mass density increase.

Now, the longitudinal and the transverse CTE of the P100 fiber are plotted (Fig. 7) as functions of the temperatures from 300 to 2500 K. Fig. 9 shows that the difference between both CTEs is higher than for other any fiber at any temperature. The values of the transverse CTE obtained for this Amoco fiber are in good agreement with the data given by [17,18].

The measurements of the fibers experimental data are summarized Tables 3 and 4 where the polynomial coefficients  $a_i$ , of the transverse and the longitudinal CTE are given with the following expression form:  $\alpha_s(T) = \sum_{i=0}^N a_i T^i$ .

### 5.3. Thermomechanical behaviour with structural and mechanical properties

In the ex-PAN fiber (HTA 5131 and PANEX 33), anisotropy (see Fig. 2) is more important than in isotropic pitch fibers (XN05). Intensity of scattering  $I(\phi)$  is quite close for ex-PAN fibers. This is the effect of the graphite layer planes orientation relative to the fiber axis. This explains the difference between the longitudinal and the transverse CTE for ex-PAN fibers. As the P100 fiber is the most anisotropic, both transverse and longitudinal CTEs will be close to the pure graphite layer plane. In order to rank such measured carbon fibers, an anisotropic parameter  $R$  between the transverse ( $\alpha_{ST}$ ) and the longitudinal ( $\alpha_{SL}$ ) specific CTE is calculated as follows:  $R = \frac{(1+\alpha_{ST})}{(1+\alpha_{SL})} - 1$ . This ratio is plotted (Fig. 8) as functions of the temperatures. This  $R$  parameter is useful for anisotropy estimation as functions of the temperatures between the transverse and the longitudinal CTEs of such carbon fibers. We clearly see that: the XN05 pitch fiber is the most isotropic, ex-PAN (HTA 5131 and PANEX 33) fibers are anisotropic and the P100 pitch fiber is the most anisotropic. This illustrates that the  $I(\phi)$  distribution is not the only parameter that could be used to find the fibers anisotropy. For ex-PAN PANEX 33 fibers, we established that HTT has an influence on properties because the anisotropy factor ( $R$ ) between the as received fiber and the HTT at 2500 K increases by two.

In order to improve relationships between the thermomechanical behaviour of the anisotropic carbon fibers with the structural ( $d_{002}$  and  $L_c$ ) and the mechanical ( $E$ ) properties we have compared the transverse and longitudinal CTEs measured at 1200 K with the other properties in Table 2 taken at room temperature. A temperature equal to 1200 K is chosen for anisotropic pitch and ex-PAN fibers evaluation. Such temperature corresponds to the maximum measured value of the HTA 5131 fiber.

In Fig. 9, the transverse and the longitudinal specific CTEs are plotted as functions of the interlayer spacing  $d_{002}$ . As expected by intensity scattering (Fig. 2), the transverse CTE increases as the interlayer spacing  $d_{002}$  decreases whereas, at the same time, the longitudinal CTE decreases. In Fig. 10, the transverse CTE increases and the longitudinal CTE decreases as the crystalline size  $L_c$  increases. These results establish that the fibers degree of ordering of crystalline and the heat treatment are directly linked to the thermomechanical values of such carbon fibers. In Fig. 11, the dependence of the CTE on the Young's modulus is demonstrated. We show that the transverse CTE increases when the Young's modulus increases. This behaviour is a result of the direct effects of the structural crystallite parameters on the mechanical, thermomechanical and thermal properties.

## 6. Conclusion

Experimental device and data processing method developed are promising for the measurements of the fibers transverse and longitudinal CTEs at very high temperatures. Amounts of metallic, ceramic and carbon fibers CTEs were measured in this study. These kinds of results allow data acquisition

in order to design and model the behaviour of composite materials at very high temperatures.

Furthermore, such measurements have also improved the influence of the carbon fibers structural properties on the thermomechanical behaviour. As measured by X-ray analysis, the most isotropic fiber (XN05) has the highest longitudinal CTE and the lowest transverse CTE. On the opposite, when the fiber is very anisotropic (such as P100), the transverse specific CTE is very high whereas the longitudinal CTE is lower. The influence of the temperature of the heat treatment is important too (see PANEX33). In fact, we observed that an increase of the temperature treatment on PAN fiber produced effects on the thermomechanical behaviour. Then, the longitudinal specific CTE decreases whereas the transverse CTE increases. Moreover, with the given anisotropic ratio  $R$ , it seems possible to rank fibers from the most isotropic to the most anisotropic. This ratio calculated as a function of the temperatures also allowed the observation of the increase of the anisotropy factor with the increasing of the temperature.

Finally, a comparison of the structural properties with the measured transverse and longitudinal CTEs allows us to demonstrate that there is a link between such properties and the thermomechanical behaviour of anisotropic carbon fibers.

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